Example 4.16

The bacteria in a culture grows by 5% in the first hour, decreases by 8% in the second hour and again increases by 10% in the third hour. Find the count of the bacteria at the end of 3 hours, if its initial count was 10000.

Solution:

Bacteria at the end of 3 hours

$$A = P\left(1+\frac{a}{100}\right)\left(1+\frac{b}{100}\right)\left(1+\frac{c}{100}\right)$$

$$= 10000\left(1+\frac{5}{100}\right)\left(1-\frac{8}{100}\right)\left(1+\frac{10}{100}\right) \text{ ('–' because 'decrease')}$$

$$= 10000 \times \frac{105}{100} \times \frac{92}{100} \times \frac{110}{100}$$

$$A = ₹10626$$

Example 4.17

The population of a town is increasing at the rate of 6% p.a. It was 238765 in the year 2018. Find the population in the year 2016 and 2020.

Solution:

Let the population in 2016 be P.

Then, $A = P\left(1+\frac{r}{100}\right)^n$

$$\Rightarrow 238765 = P\left(1+\frac{6}{100}\right)^2 = P\left(\frac{53}{50}\right)^2$$

$$\Rightarrow P = 238765 \times \frac{50}{53} \times \frac{50}{53}$$

$$\therefore P = 212500$$

Let the population in 2020 be A

Then, $A = P\left(1+\frac{r}{100}\right)^n$

$$\therefore A = 238765\left(1+\frac{6}{100}\right)^2$$

$$= 238765 \times \frac{53}{50} \times \frac{53}{50}$$

$$= 95.506 \times 53 \times 53$$

$$A = 268276$$

$\therefore$ The population in the year 2016 was 212500 and that in the year 2020 will be 268276.

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